

B.Tech CSE subject-wise revision map

Built from your CIL MT Paper II response sheet (24 Aug 2026). Accuracy figures are your actual performance on the 100 domain questions.

Target exams: HAL MT (5-6 Sep) · IOCL Grade A (24 Sep) · IBPS SO IT Officer Mains (1 Nov)

Priority order

#	Subject	Your CIL accuracy	Tested in
1	Computer Organization & Architecture	6/17 (35%)	CIL, IOCL, HAL, IBPS
2	DBMS	3/8 (37%)	CIL, IOCL, HAL, IBPS
3	Operating Systems	8/14 (57%)	CIL, IOCL, HAL, IBPS
4	Computer Networks	6/11 (54%)	CIL, IOCL, HAL, IBPS
5	Data Structures & Algorithms	15/19 (78%)	CIL, IOCL, HAL, IBPS
6	Theory of Computation	6/12 (50%)	CIL, IOCL, HAL — not in IBPS
7	Compiler Design	2/9 (22%)	CIL, IOCL, HAL — not in IBPS
8	Discrete Maths / Probability	3/7 (42%)	CIL, IOCL, HAL — light in IBPS
9	Programming in C	1/1 (100%)	CIL, IOCL, HAL, IBPS (as OOP too)
—	Software Engineering	not tested at CIL	IBPS only — not in CIL, IOCL or HAL
—	Object-Oriented Programming	not tested at CIL	IBPS only — light elsewhere
—	Artificial Intelligence / Machine Learning	not tested at CIL	interview stage
—	Python	not tested at CIL	interview stage

Priority is set by weakness multiplied by how many of your four exams test the subject. Computer Organization is first because it is your worst subject and the only one that appears in all four papers.

1. Computer Organization & Architecture

Your CIL score: 6/17 · Tested in: CIL, IOCL, HAL, IBPS

Topics to cover

Number systems and base conversion — binary, octal, hex, 2s complement

Powers of two: $2^{10}=1\text{K}$, $2^{20}=1\text{M}$, $2^{30}=1\text{G}$. Address bits to memory size

IEEE 754 single and double precision — sign, biased exponent, mantissa

Memory hierarchy order: registers, cache, main memory, secondary storage

Cache mapping: direct, associative, set-associative; tag/index/offset split

Associative memory: argument register, key register, match register, memory array logic

Instruction formats: zero, one, two, three address; addressing modes

I/O organisation: programmed I/O, interrupt-driven, DMA

Secondary storage: platter, track, sector, cylinder; seek + rotational + transfer time

RAID 0 striping, RAID 1 mirroring, RAID 5 distributed parity

SRAM vs DRAM; refresh, speed, cost, use

Digital logic: K-maps, flip-flops (SR, JK, D, T), counters, registers

Sequential circuits: state diagram, state table, Mealy vs Moore

What you actually got wrong at CIL

Q22 — Which of the following is the correct symbol for representing the decimal number 12 in hexadecimal system?

You marked: D

Correct: C

Q31 — In an associative memory system, the argument register is used to:

You marked: Hold data to be written into memory

Correct: Compare with key fields in all memory locations simultaneously

Q39 — A 32-bit IEEE 754 single-precision floating-point number is represented as: 0100 0001 0100 1001 0000 0000 0000 0000 What is the decimal value of this floating-point number?

You marked: 100.5

Correct: 12.5625

Q43 — In an associative memory system, the key register holds:

You marked: The memory address of the required data.

Correct: The tag field used for parallel comparison.

Q46 — Which of the following best describes the memory hierarchy in a computer system?

You marked: Secondary Storage → Main Memory → Cache → Registers

Correct: Registers → Cache → Main Memory → Secondary Storage

Q48 — During a content search operation in associative memory, which register temporarily holds the key used to compare against all stored memory words simultaneously?

You marked: Program Counter

Correct: Argument Register

Q61 — Which statement best describes a zero-address instruction in stack-based architectures?

You marked: Operands are fetched directly from main memory.

Correct: Operands are implicitly taken from the top of the stack.

Q74 — Which statement correctly differentiates between SRAM and DRAM?

You marked: DRAM is faster and more expensive than SRAM

Correct: SRAM is faster but more expensive than DRAM

Q75 — What is the output of a JK flip-flop when both inputs $J = 1$ and $K = 1$?

You marked: Output is set to 0

Correct: Output toggles

Q93 — Which of the following best describes the interaction between CPU and DMA during block transfer?

You marked: CPU transfers data to cache, DMA transfers it to I/O

Correct: CPU sends control information once, then DMA handles the transfer independently

Q99 — The hardware organization of associative memory consists of:

You marked: Argument Register(A), Memory Array Logic and Match Register(M) Only

Correct: Memory Array Logic, Argument Register(A), Key Register(K) and Match Register(M)

2. DBMS

Your CIL score: 3/8 · Tested in: CIL, IOCL, HAL, IBPS

Topics to cover

Relational model vocabulary: relation, tuple, attribute, domain, degree, cardinality

Keys: super, candidate, primary, alternate, foreign, composite

Integrity constraints: entity integrity (no NULL primary key), referential integrity

ER diagram symbols: rectangle = entity, ellipse = attribute, diamond = relationship

Relational algebra: select (unary, horizontal), project (unary, vertical), join, division

Normalisation: 1NF through BCNF, functional dependencies, decomposition

SQL: DDL vs DML, joins, aggregate functions, group by / having, subqueries

Transactions and ACID; concurrency control; isolation levels

Recovery: redo log, undo log, write-ahead logging, checkpoints

Indexing: B-tree, B+ tree, hashing; storage and file structure (RAID again)

What you actually got wrong at CIL

Q5 — Identify the correct option for project operation in relational algebra: Statement-1: The project operation is a binary operation that returns its argument relation, with certain tuple left out. Statement-2: The result of the PROJECT operation can be visualized as a horizontal partition of the relation into two relations: one has the needed rows (tuples) and contains the result of the operation, and the other contains the discarded rows.

You marked: Statement-1 is True and Statement-2 is True

Correct: Statement-1 is False and Statement-2 is False

Q25 — What does each row in a relational table represent?

You marked: Attribute

Correct: Tuple

Q56 — What shape is commonly used to represent an entity in an ER diagram?

You marked: Ellipse

Correct: Rectangle

Q68 — The entity integrity constraint ensures that:

You marked: All attribute values must belong to the same domain

Correct: No primary key value can be NULL

Q96 — A banking transaction commits successfully, but a system crash occurs immediately afterward. Which mechanism ensures durability?

You marked: Transaction isolation

Correct: Redo log

3. Operating Systems

Your CIL score: 8/14 · Tested in: CIL, IOCL, HAL, IBPS

Topics to cover

Process vs thread; process states; PCB; context switching

System calls: read, write, fork, exec, wait — kernel space vs user space

CPU scheduling: FCFS, SJF, SRTF, priority, round robin; preemptive vs non-preemptive

Turnaround, waiting, response time calculations from Gantt charts

Synchronisation: critical section, semaphores, mutex, producer-consumer, readers-writers

Deadlock: four conditions, prevention, avoidance (bankers algorithm), detection

Memory management: fixed and variable partitioning, internal vs external fragmentation

Paging, segmentation, page table, TLB, effective access time

Virtual memory and demand paging; page faults

Page replacement: FIFO, LRU, Optimal, clock; Beladys anomaly (FIFO only)

Thrashing and the working set model

Disk scheduling: FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK

File systems: allocation methods, directory structure, free space management

What you actually got wrong at CIL

Q1 — What does the read() system call in an operating system do?

You marked: Copies data directly from device hardware into user space without kernel intervention

Correct: Copies data from kernel space to user space, possibly after reading from device buffers

Q49 — What is the division of the main memory in Fixed Partitioning in memory management of operating system?

You marked: Created as needed by the processes.

Correct: Made at boot time and remain unchanged in size and number.

Q63 — Which of the following Windows API functions is used to write data to a file?

You marked: fwrite()

Correct: WriteFile()

Q66 — You are tasked to construct a binary search tree (BST) from a stream of incoming sensor readings, where each reading is processed as it arrives. Midway, the sensor malfunctions and emits a duplicate reading not originally in the stream. To preserve BST properties and ensure efficient future search operations, which strategy should you employ for the duplicate reading?

You marked: Allow the duplicate in the left subtree and continue

Correct: Ignore the duplicate and continue inserting unique values

Q82 — Which of the following statements correctly describes Isolation and Transaction Isolation Levels in a database system? Statement-1: Ensuring the isolation property is the responsibility of a component of the database system called the concurrency-control system. Statement-2: The isolation levels may be Serializable, Repeatable read, Read committed, Read uncommitted and all the isolation levels additionally disallow dirty writes, that is, they disallow writes to a data item that has already been written by another transaction that has not yet committed or aborted.

You marked: Statement-1 is True and Statement-2 is False

Correct: Statement-1 is True and Statement-2 is True

Q89 — Why is it considered good practice to close files explicitly even if the programming language runtime may close them at program termination?

You marked: To guarantee file permissions are reset

Correct: To release resources promptly and avoid resource leaks during execution

4. Computer Networks

Your CIL score: 6/11 · Tested in: CIL, IOCL, HAL, IBPS

Topics to cover

OSI seven layers and TCP/IP model — function of each layer
Layered design issues: addressing, error control, flow control, sequencing
Connection-oriented vs connectionless; service vs protocol
Physical layer: transmission media, topologies, multiplexing
Data link: framing, error detection (CRC, checksum, Hamming), ARQ protocols
MAC: CSMA/CD, CSMA/CA, Ethernet, switching
Network layer: IPv4 addressing, classes, subnetting, CIDR, NAT; routing algorithms
Transport layer: TCP vs UDP, three-way handshake, flow and congestion control, sliding window
Application layer: DNS hierarchy (root, TLD, authoritative), HTTP, FTP, SMTP, DHCP
Network devices: hub, switch, router, gateway, bridge

What you actually got wrong at CIL

Q24 — Which of the following statements correctly differentiates connection- oriented from connectionless services?

You marked: Connectionless has higher overhead; connection-oriented has minimal.

Correct: Connection-oriented ensures reliability and order; connectionless may not.

Q30 — What do you mean by TLD name servers?

You marked: Tiered-level domain name servers

Correct: Top-level domain name servers

Q36 — Which of the following statements best reflects a comprehensive layered design issue in computer networks?

You marked: The design issues of addressing and error control are handled exclusively by the network

Correct: Each layer must handle key issues like addressing, error control, flow control, sequencing

Q85 — Which one of the following statements is true for a Hermitian matrix?

You marked: All entries are real numbers

Correct: All diagonal elements are real numbers

Q94 — Which scenario is least suitable for sparse matrix representation in computer memory?

You marked: A diagonal matrix

Correct: A matrix where most elements are nonzero

5. Data Structures & Algorithms

Your CIL score: 15/19 · Tested in: CIL, IOCL, HAL, IBPS

Topics to cover

Arrays, linked lists (singly = data + one pointer, doubly = data + two pointers), circular lists

Stack and queue; circular queue rear/front updates with wraparound

Trees: binary tree, BST, AVL, heap; traversals and reconstruction rules

Preorder + Inorder reconstructs a unique tree; Postorder + Inorder does too; Preorder + Postorder does not

Graphs: representation, BFS, DFS, shortest path, MST

Hashing: hash functions, collision resolution, load factor

Sorting: bubble, insertion, selection, merge, quick, heap — best/average/worst complexity

Searching: linear, binary; time and space complexity notation

Algorithm design: divide and conquer, greedy, dynamic programming

What you actually got wrong at CIL

Q26 — Which of the following traversals can be used to reconstruct a unique binary tree, if combined with Preorder?

You marked: Postorder

Correct: Inorder

Q33 — What is the best approach to delete an element from an unsorted array?

You marked: Sort the array and then delete

Correct: Replace the element with the last element and reduce the array size

Q38 — In a circular queue implemented using an array, how is the rear counter updated after an enqueue operation?

You marked: No update to rear pointer is needed in the enqueue operation

Correct: Incremented and wrapped around to the start if needed

Q59 — What does a node in a singly linked list typically contain?

You marked: Data and one pointer

Correct: Data and two pointers

6. Theory of Computation

Your CIL score: 6/12 · Tested in: CIL, IOCL, HAL — not in IBPS

Topics to cover

Finite automata: DFA, NFA, epsilon-NFA, conversions, minimisation

Regular expressions: a^* means zero or more, a^+ means one or more

Constructing regex from a language description and vice versa

Mealy and Moore machines; finite state transducers; conversion between them

Regular vs context-free languages; pumping lemma

Context-free grammars, derivations, parse trees, ambiguity

Pushdown automata; Turing machines; decidability basics

What you actually got wrong at CIL

Q4 — Consider an FSM that accepts all strings over $\{a, b\}$ with an even number of a's. Which regular expression corresponds to it?

You marked: $(a+b)^*a(a+b)^*(b+c)$

Correct: $(b^*(ab^*ab^*))^*$

Q15 — Consider a finite state transducer (FST) with $Q = \{q_0, q_1\}$, $\Sigma = \{0, 1\}$, $\Gamma = \{a, b, c\}$, initial state q_0 , and the following transitions: $\delta(q_0, 0) = q_1$, $\delta(q_0, 1) = q_0$, $\delta(q_1, 0) = q_0$, $\delta(q_1, 1) = q_1$ and output function: $\theta(q_0, 0) = a$, $\theta(q_0, 1) = c$, $\theta(q_1, 0) = b$, $\theta(q_1, 1) = a$. What will be the output string for the input 1010?

You marked: caba

Correct: caab

Q17 — A system must recognize any number of consecutive 'a's. Which regular expression correctly models this?

You marked: a^+

Correct: a^*

Q21 — Consider a Mealy machine designed to process a sequence representing the presence of mineral samples ('M') and non-mineral samples ('N') during a geological survey. The machine outputs 'A' when the current sample is 'M' and the previous sample was 'N', otherwise it outputs 'B'. Which description below correctly reflects the machine's output behavior?

You marked: It changes output only when the input switches from 'M' to 'N'.

Correct: It produces output based on both the current input and the previous state.

Q40 — Which of the following correctly describes an SQL schema?

You marked: An SQL schema is a view that aggregates data from multiple tables.

Correct: An SQL schema is identified by a name, includes an authorization identifier for its owner,

Q98 — Which statement correctly defines the regular expressions? Statement-1: Regular expressions representation doesn't involve a combination of strings of symbols from some alphabet Σ , parentheses and the operators $+$, $\times$, and $*$. Statement-2: A regular expression is not obtained from the symbol $\{a, b, c\}$, empty string ϵ , and empty-set Φ perform the operations $+$, $\times$ and $*$ (union, concatenation and Kleene star).

You marked: Statement-1 is False and Statement-2 is True

Correct: Statement-1 is False and Statement-2 is False

7. Compiler Design

Your CIL score: 2/9 · Tested in: CIL, IOCL, HAL — not in IBPS

Topics to cover

Phases of a compiler: lexical, syntax, semantic, intermediate code, optimisation, code generation

Lexical analysis: tokens, lexemes, patterns; DFA-based recognition of identifiers

Top-down parsing: recursive descent, predictive LL(1), FIRST and FOLLOW sets

Left recursion (direct and indirect) and left factoring — why LL(1) needs them removed

Bottom-up parsing: shift-reduce; the four actions are shift, reduce, accept, error

LR parsers: SLR, CLR, LALR

Symbol table; static vs dynamic scope; lexical scoping

Code optimisation techniques

What you actually got wrong at CIL

Q29 — Which parsing method uses a stack to store symbols and reduces them when a right-hand side pattern is recognized?

You marked: Top-Down Parsing

Correct: Shift-Reduce Parsing

Q35 — In a shift-reduce parser, which of the following lists includes all four possible parser actions?

You marked: Reduce, Copy, Accept, Abort

Correct: Shift, Reduce, Accept, Error

Q58 — Consider the following grammar $G: E \rightarrow T E' E' \rightarrow + T E' \varepsilon T \rightarrow F T' T' \rightarrow * F T' \varepsilon F \rightarrow (E)$ id Which of the following statements about top-down parsing of grammar G is true?

You marked: The grammar requires backtracking since FIRST and FOLLOW sets of E' overlap.

Correct: The grammar is suitable for LL(1) (predictive) parsing and does not require backtracking.

Q60 — Which of the following is an application of compiler technology?

You marked: Cache mapping

Correct: Code optimization

Q62 — _____ is NOT a valid token typically generated by a lexical analyzer.

You marked: Compiler

Correct: Parser

Q83 — Which statement about left recursion in context-free grammars is accurate?

You marked: Left recursion is required for building LL(1) parsers without any modification.

Correct: Left recursion can be direct or indirect and may lead to infinite recursion in top-down

Q87 — In the context of programming languages, if a compiler can determine the scope of a declaration by examining the program text alone, the language is said to use:

You marked: Static policy and dynamic scope

Correct: Static policy and lexical scope

8. Discrete Maths / Probability

Your CIL score: 3/7 · Tested in: CIL, IOCL, HAL — light in IBPS

Topics to cover

Set theory: subsets, power set (2^n subsets), union, intersection, complement

Relations and functions; equivalence relations; partial orders

Propositional and predicate logic

Combinatorics: permutations, combinations, pigeonhole

Graph theory basics

Probability: conditional probability, Bayes theorem

Random variables: discrete and continuous, expectation, variance, covariance = $E(XY) - E(X)E(Y)$

Matrices: determinant, cofactor expansion, rank, eigenvalues; Hermitian and symmetric matrices

What you actually got wrong at CIL

Q53 — Which of the following correctly represents the cofactor expansion of a 3x3 determinant along the first row?

You marked: Sum of elements of the first column times their cofactors

Correct: Sum of each element of the first row times its corresponding cofactor

Q54 — If a measurement is recorded as 12.8 cm and the true value is 13.0 cm, what is the absolute error?

You marked: 0 cm

Correct: 0.2 cm

Q64 — Given two random variables X and Y with $E(X) = 2$, $E(Y) = 3$, and $E(XY) = 10$, what is the covariance of X and Y?

You marked: 1

Correct: 4

Q80 — Which of the following statements correctly describes the power set of the set {a, b}?

You marked: It is the set containing only the singleton subsets of {a, b}

Correct: It is the set containing all possible subsets of {a, b}

9. Programming in C

Your CIL score: 1/1 · Tested in: CIL, IOCL, HAL, IBPS (as OOP too)

Topics to cover

Data types, storage classes, operators and precedence

Pointers: declaration, dereferencing, pointer to pointer, pointer arithmetic

Arrays and pointers: 1D array name is a pointer to the first element; 2D array declaration

Functions, call by value vs call by reference, recursion

Structures, unions, dynamic memory (malloc, calloc, free)

File handling: open, read, write, close — why explicit close matters

Additional subjects for IBPS SO only

Neither of these appeared on your CIL paper and neither is in the GATE-pattern syllabus used by IOCL and HAL. Leave them until October, then cover them properly — the questions are almost entirely definitional, which makes them some of the cheapest marks in the IBPS paper.

Software Engineering

IBPS only — not in CIL, IOCL or HAL

SDLC models: waterfall, spiral, V-model, incremental, agile and scrum

Requirement analysis and SRS

Design concepts: cohesion (want high) and coupling (want low), modularity

Testing: unit, integration, system, acceptance; black box vs white box; alpha vs beta

Software metrics, COCOMO, risk management

Maintenance types: corrective, adaptive, perfective, preventive

Object-Oriented Programming

IBPS only — light elsewhere

Four pillars: encapsulation, abstraction, inheritance, polymorphism

Class, object, constructor, destructor

Overloading (compile time) vs overriding (run time)

Types of inheritance; the diamond problem; virtual functions

Access specifiers: public, private, protected

Abstract classes and interfaces

AI and Python modules

Worth being clear about where these pay off. Neither subject appears in the CIL, IOCL, HAL or IBPS professional knowledge syllabus, and CIL asked its two programming questions in C. So for the written papers, these are zero-weight.

Where they matter is the stage after. IOCL selection is CBT (85%), group discussion and group task (5%), then personal interview (10%). HAL is written test (85%) plus interview (15%). IBPS has an interview round too. In those rooms, your AI project and your Python work are what you will actually be asked about — and they are the part of your profile that separates you from everyone else who also knows what a page fault is. Prepare them as interview material, not as exam syllabus.

Artificial Intelligence / Machine Learning

Not in the CIL, IOCL, HAL or IBPS professional knowledge syllabus. Value is in the interview stage.

Supervised vs unsupervised vs reinforcement learning

Classification vs regression; training, validation, test split

Overfitting and underfitting; bias-variance tradeoff; regularisation

Common algorithms: linear and logistic regression, decision trees, random forest, KNN, SVM, k-means, naive Bayes

Evaluation: accuracy, precision, recall, F1, confusion matrix

Neural network basics: perceptron, activation functions, backpropagation, gradient descent

CNN for images, RNN and LSTM for sequences, transformers and attention at a high level

Search in classical AI: BFS, DFS, A*, hill climbing, minimax and alpha-beta pruning

Knowledge representation, expert systems, fuzzy logic

Be ready to explain your own AI project end to end — dataset, model choice, metric, result, limitation

Python

Not tested as a coding subject in these four exams — CIL asked C, and GATE-pattern papers use C. Interview and project value only.

Data types and mutability: list, tuple, set, dict, string

List comprehensions, slicing, unpacking

Functions, *args and **kwargs, lambda, map, filter, reduce

OOP in Python: classes, self, **init**, inheritance, dunder methods

Exception handling, file I/O, with statement and context managers

Modules and packages, virtual environments, pip

Standard libraries: numpy arrays and broadcasting, pandas DataFrame operations, matplotlib plotting

scikit-learn workflow: fit, predict, train_test_split, pipelines

Decorators, generators and iterators, GIL — asked in technical interviews